Supplementary Information

A Coupled Chemo-Mechanical and Life-Cycle Framework for Oyster-Shell-Derived Cementitious Materials in Aggressive Infrastructure Environments

Tables S1–S4 give the full calibration and re-evaluation results summarised in Sections 4.4 and 7.4 of the manuscript. All numbers are generated by calibrate_abbas.py, kf0_prediction.py and ShellConcreteMC_v3.py in the repository https://github.com/sandlerleon/oyster-shell-concrete-mc.

Table S1. Calibration set-up

Item	Value
Dataset	Abbas and Kudukkan (2025), Constr. Mater. 5(4):82, scallop-shell powder, binder 350 kg/m3, water 160 kg/m3, 28-day strength
Mixes (shell-derived powder)	S0F15 50 kg/m3 (r=0.143, d=90 um); S15F15 95 (r=0.271, d=64); S30F15 140 (r=0.400, d=55)
Observed relative strength	S0F15 1.00 (text: 'almost the same'); S15F15 0.865 (derived from reported moduli, E ~ sqrt(fc)); S30F15 0.75 (text: 'nearly 25% reduction')
Extension	filler efficiency k_f ~ U(0, 1.2); cement volume in the gel/space ratio multiplied by 1 + k_f r/(1 - r)
Algorithm	rejection ABC; 300,000 prior draws per case; accept if | relative strength - observed| <= 0.07 for S0F15 and S30F15; S15F15 held out
Accepted draws	10999 (A), 10975 (B), 10999 (C); acceptance 3.7%
Held-out mix within tolerance	100% of accepted draws (interpolation between the calibration mixes)
Original model (k_f = 0)	predicted relative strength S0F15/S15F15/S30F15: 0.79, 0.61, 0.43; none of 100,000 draws within tolerance

Table S2. Prior and posterior parameter summaries (case A draws; cases B and C are identical to within sampling error)

Parameter	Prior median	Posterior median	Posterior P10	Posterior P90
kf	0.599	0.648	0.537	0.839
nu	0.75	0.71	0.13	1.34
bpack	0.15	0.169	0.0378	0.276
o_org	0.0213	0.0218	0.0128	0.0322
eta_w	0.6	0.579	0.354	0.827
kap_o	31.6	41.3	7.92	153
kap_s	2.4	2.31	1.89	2.84
kimp	0.025	0.0248	0.00478	0.045
k0	0.26	0.259	0.2	0.334

acap	0.935	0.936	0.907	0.963
S0	78	78.1	68.7	88.9

Table S3. Prior (original model, k_f = 0) and calibrated results for P1, 30 um shell, all cases and designs

Case	Design	dE P50 (%) orig/cal	fc28 vs OPC orig/cal	Life P50 yr orig/cal	OPC life P50 orig/cal	P(non-inferior) orig/cal	P(adv, mat) orig/cal
A	10% same w/b	-8.5 / -8.5	0.87 / 0.97	21.2 / 22.3	25.1 / 24.6	0.05 / 0.27	0.05 / 0.27
A	25% same w/b	-21.2 / -21.2	0.66 / 0.91	14.9 / 18.5	25.1 / 24.6	0.01 / 0.23	0.01 / 0.23
A	25% matched	-0.3 / -16.4	1.00 / 1.00	30.8 / 21.3	25.1 / 24.6	0.91 / 0.28	0.01 / 0.28
A	50% same w/b	-42.5 / -42.5	0.33 / 0.77	0.1 / 11.8	25.1 / 24.6	0.00 / 0.18	0.00 / 0.18
A	50% matched	+3.5 / -33.1	0.99 / 1.00	43.5 / 16.1	25.1 / 24.6	0.92 / 0.24	0.01 / 0.24
B	10% same w/b	-8.1 / -8.1	0.86 / 0.96	100.0 / 100.0	100.0 / 100.0	0.56 / 0.55	0.56 / 0.55
B	25% same w/b	-20.2 / -20.2	0.64 / 0.90	100.0 / 100.0	100.0 / 100.0	0.05 / 0.04	0.05 / 0.04
B	25% matched	-0.1 / -15.4	1.00 / 1.00	100.0 / 100.0	100.0 / 100.0	0.98 / 0.19	0.00 / 0.19
B	50% same w/b	-40.4 / -40.5	0.30 / 0.75	80.2 / 80.3	100.0 / 100.0	0.00 / 0.00	0.00 / 0.00
B	50% matched	+3.7 / -31.2	1.00 / 1.00	100.0 / 100.0	100.0 / 100.0	0.99 / 0.02	0.01 / 0.02
C	10% same w/b	-8.5 / -8.5	0.87 / 0.97	15.7 / 15.6	18.2 / 18.1	0.00 / 0.00	0.00 / 0.00
C	25% same w/b	-21.3 / -21.4	0.66 / 0.91	12.7 / 12.6	18.2 / 18.1	0.00 / 0.00	0.00 / 0.00
C	25% matched	-0.3 / -16.5	1.00 / 1.00	21.4 / 14.0	18.2 / 18.1	0.93 / 0.00	0.01 / 0.00
C	50% same w/b	-42.7 / -42.7	0.33 / 0.77	8.8 / 9.1	18.2 / 18.1	0.00 / 0.00	0.00 / 0.00
C	50% matched	+3.5 / -33.2	0.99 / 1.00	28.7 / 10.7	18.2 / 18.1	0.97 / 0.00	0.01 / 0.00

Table S4. Carbon check against the same paper

Item	Value
Control embodied carbon (paper's factors)	331.0 kg CO2e/m3
S15F15 / S30F15 change vs control	-24.4% / -36.0%
Model shell emission factor at 55 um (P10 / P50 / P90)	0.036 / 0.058 / 0.084 kg CO2e/kg
Laboratory value in the paper	0.059 kg CO2e/kg (system boundary not stated)

The first two rows use the same budget arithmetic as the model and are a consistency check, not an independent validation.